No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	SWTID-2026-2085
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Soil Data Input	Allows farmers to enter soil nutrients (N, P, K, pH) and environmental data.	User Interface	Done	High
2	Data Validation	Validates user inputs before processing.	Backend	Done	
3	Crop Recommendation	Predicts the most suitable crop using the trained machine learning model.	ML Prediction Module	Done	High
4	Machine Learning Model	Trained classification model for accurate crop prediction	ML Model	Done	High
5	Result Display	Displays the recommended crop with prediction results.	Frontend	Done	
6	Dataset Management	Uses the Crop Recommendation Dataset for training and prediction.	Data Layer	Done	
7	Model Loading	Loads the trained model (model.pkl) during application startup.	backend	Done	
8	Web Application	Provides an easy-to-use web interface for farmers.	Flask Application	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Soil Data Input, Result Display
2	Backend / Logic	3	Data Validation, Model Loading, Prediction Processing
3	Database / Storage	1	Crop Recommendation Dataset (CSV), Trained Model (.pkl)
4	API / Integration	1	Flask Application Routing
5	Security / Authentication	1	Input Validation and Error Handling